

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: **SYNQ**

Project / Product Name: **AVY — Streaming Live RAG Assistant**

Organization / Institution (if any): **SRM Institute of Science and Technology**

Submission Date: **27th September, 2026**

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? Yes / No

YES

AI tools were used as development-assistance tools during the design, implementation, debugging, testing, documentation, and refinement of AVY.

The team retained responsibility for the final system architecture, implementation decisions, integration, testing, verification, and submission. AI-generated suggestions and code were reviewed and adapted before being incorporated into the project.

AI was **not treated as an autonomous source of truth**. Project functionality, benchmark results, latency measurements, retrieval results, citations, and test results were required to come from the actual implemented and tested system.

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming

AI was used during the early design stage to explore approaches for building a streaming conversational RAG system capable of handling incomplete speech transcripts, multiple intents, follow-up questions, corrections, and late-arriving information.

AI assistance was used to discuss and refine:

- Streaming retrieval architecture
- Retrieval gating
- WAIT / RETRIEVE / NO_RETRIEVE states
- Early retrieval
- Multi-intent query decomposition
- Multi-query retrieval

- Evidence fusion
- Reranking
- Session-aware refinement
- Grounded response generation
- Citation traceability
- Streaming synthesis
- Telemetry and evaluation

The final architecture was selected and adapted by the team according to the requirements of Samsung PRISM Theme 04.

Code generation or assistance

AI was used as a coding assistant for selected implementation tasks, including:

- Generating implementation suggestions
- Producing code scaffolding
- Explaining unfamiliar APIs and programming concepts
- Reviewing functions and modules
- Identifying potential implementation errors
- Suggesting refactoring approaches
- Assisting with integration between system components
- Generating test-case ideas

AI-generated code was not blindly accepted. The team reviewed, modified, integrated, tested, and debugged the resulting implementation.

UI / UX design

AI assistance was used to brainstorm and refine the demonstration interface for AVY.

The UI was designed to make the project's core Streaming Live RAG behaviour visible to hackathon judges, including:

- Live transcript
- Current retrieval state
- WAIT / RETRIEVE / NO_RETRIEVE status
- Generated retrieval queries

- Retrieved evidence
- Source information
- Grounded answer
- Citations
- Streaming response
- Performance and latency information

The team selected and implemented the final interface according to the actual system capabilities.

Content creation

AI was used to assist with:

- README preparation
- Technical documentation
- Architecture explanations
- Feature descriptions
- Setup instructions
- Demo scenario descriptions
- Code documentation
- Project presentation/supporting material

The team reviewed the generated content and ensured that documentation reflected the actual implemented system.

Data analysis

AI assistance was used to help interpret and organize evaluation-related information and identify useful metrics for the system.

The project evaluates areas including:

- Retrieval Recall
- Answer Groundedness
- Time-to-First-Token
- Cost per Turn
- Retrieval-controller decisions

- Query decomposition
- Retrieval latency
- Fusion and reranking
- Session refinement
- Citation generation
- End-to-end latency

No evaluation results were generated or invented by AI. Reported results are intended to come from actual execution of the implementation.

Testing / debugging

AI was used as a debugging and testing assistant to:

- Identify possible bugs
- Analyze error messages
- Suggest debugging approaches
- Generate test cases
- Identify edge cases
- Review component interactions
- Suggest improvements to error handling
- Assist with testing streaming and retrieval behaviour

Final verification was performed against the actual implementation.

Other

AI was also used as a technical research and development assistant to understand the relationship between the existing AVI architecture and the new AVY architecture.

The project brief specifies that AVY is a **new repository** derived from relevant architectural concepts from AVI rather than a direct copy of AVI.

AI assistance was therefore used to help analyze which architectural concepts could be adapted while keeping the AVY implementation focused on Streaming Live RAG

4. Feature Origin Classification (Please add/delete based on the number of the feature)

1. Feature Name: **Retrieval Controller — WAIT / RETRIEVE / NO_RETRIEVE**
2. Self-Generated/AI-Generated/Both _: **Both — Self-Generated + AI-Assisted**
3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

OpenAI

Purpose of AI Usage:

Architecture exploration, implementation assistance, edge-case analysis, and debugging.

Representative Prompt Used:

“Design a retrieval controller for a streaming conversational RAG system that determines whether an incomplete transcript should WAIT, whether sufficient information exists to RETRIEVE, or whether retrieval is unnecessary.”

AI Output Summary:

AI suggested a three-state retrieval-control architecture based on semantic completeness, information need, query quality, session context, and whether additional transcript chunks could materially change the retrieval intent.

Team Modification / Contribution:

The team adapted the proposed approach to AVY's streaming architecture and project requirements. The final controller was implemented as part of AVY and tested against WAIT, RETRIEVE, and NO_RETRIEVE scenarios.

The project specification explicitly requires these three-controller states

1. Feature Name: **Streaming Early Retrieval**
2. Self-Generated/AI-Generated/Both: **Both — Self-Generated + AI-Assisted**
3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

OpenAI

Purpose of AI Usage:

Architecture design and implementation reasoning.

Representative Prompt Used:

“How can a streaming RAG system begin retrieval before the user has finished speaking while still allowing later transcript chunks to refine the original query?”

AI Output Summary:

AI suggested treating transcript chunks as incremental information and triggering retrieval once sufficient semantic information becomes available, while allowing subsequent chunks to refine the query.

Team Modification / Contribution:

The team incorporated the approach into AVY's retrieval-controller and session-refinement pipeline. The implementation was designed around the project scenario in which an initial chunk may produce WAIT, a later chunk may trigger RETRIEVE, and subsequent information may trigger refinement.

This behaviour is a core requirement of the project's Streaming Live RAG design.

1. Feature Name: **Evidence Fusion & Reranking**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description:

AI Tool/Platform Used: OpenAI

Prompt Used:

"Design an evidence fusion and reranking pipeline for a multi-query RAG system that combines results from multiple retrieval queries, removes duplicate evidence, and ranks the most relevant results."

Output Summary:

AI suggested a pipeline involving result normalization, deduplication, Reciprocal Rank Fusion (RRF), and a subsequent reranking stage to identify the most relevant evidence.

Modification:

The team adapted the suggested approach to AVY's retrieval architecture and integrated evidence from multiple subqueries while preserving source and document metadata. The reranking stage was kept configurable to support different latency and testing requirements.

1. Feature Name: **Session-Aware Context & Refinement**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description:

AI Tool/Platform Used: OpenAI

Prompt Used:

"Design a session-aware conversational RAG system that can understand follow-up questions and refine previous retrieval when the user provides additional information."

Output Summary:

AI suggested maintaining session-level conversation turns, previous user intent, previous

subqueries, retrieved evidence, and previous answers so that follow-up requests could be interpreted in context.

Modification:

The team adapted the approach to AVY's session architecture. The system uses previous conversational context when processing follow-up questions and can update retrieval when new information changes or refines the original intent. Session context remains scoped to the current conversation.

1. Feature Name: **Grounded Streaming Answers & Citations**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description:

AI Tool/Platform Used: OpenAI

Prompt Used:

Design a grounded RAG response pipeline that generates answers from retrieved evidence, maintains source traceability, provides citations, and streams the final response."

Output Summary:

AI suggested providing the language model with the user's intent, session context, retrieved evidence, and source metadata. It also suggested maintaining document and evidence information throughout the pipeline so that citations could be traced to the retrieved sources.

Modification:

The team integrated the approach into AVY's grounding and streaming pipeline. Retrieved evidence and source metadata are used during response generation, and the system is designed to avoid fabricated facts or citations. The final response is streamed through the project's provider/gateway architecture.

1. Feature Name: **Telemetry, Evaluation & Demonstration UI**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description:

AI Tool/Platform Used: OpenAI

Prompt Used:

"Design telemetry, evaluation, testing, and a hackathon demonstration interface for a streaming RAG system that clearly shows retrieval states, retrieved evidence, citations, streaming responses, and performance metrics."

Output Summary:

AI suggested structured telemetry events, evaluation metrics such as retrieval quality, groundedness and time-to-first-token, test scenarios for the major RAG components, and a UI displaying the live transcript, retrieval state, queries, evidence, answer, citations, and latency.

Modification:

The team adapted these suggestions to the actual AVY implementation. Telemetry was structured around the pipeline stages, evaluation and testing were aligned with the implemented components, and the demonstration UI was designed to make AVY's Streaming Live RAG behavior understandable.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I AGREE**

Additional Declaration

AI was used as an assistant and development-support tool and not as a replacement for the team's technical responsibility.

The team reviewed and adapted AI-assisted outputs before integration.

The team remains responsible for:

- Final architecture
- Source-code integration
- System behaviour
- Testing
- Evaluation
- Documentation
- Demonstration
- Final submission

No intentionally fabricated:

- Benchmark results
- Retrieval results
- Citations
- Latency measurements
- Telemetry

- Test results
- System capabilities

are represented as actual project results.

The implementation is treated as the source of truth.

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: **K. SRIVIDYA**

Role: Team Representative

Signature

A handwritten signature in black ink that reads "K. Srividya". The script is cursive and fluid, with the first letter 'K' being prominent and the last letter 'a' having a long, sweeping tail.

Date: 27th September 2026